# Yucheng (Cecilia) Zhang

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## Education

**Northwestern University**, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL
**The University of Texas at Austin**, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX

## Technical Skills

**ML/AI:** Scikit-learn, PyTorch, TensorFlow, classification, feature engineering, model evaluation, ROC/AUC, backtesting
**LLM Systems:** OpenAI, Gemini, Claude, Large Language Models (LLMs), LangChain, CrewAI, prompt design, context filtering, LLM evaluation
**Data Science:** Pandas, NumPy, R, MATLAB, MNE, ggplot2, statistics, permutation-based FDR, bioinformatics, EEG/fNIRS
**Data/Cloud:** SQL Server, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI, Linux

## Machine Learning, AI, and Data Experience

### CodeAssist – Full-Stack AI Feedback System | GitHub — Aug. 2025 – Present

*University of Texas at Austin | Flask, React, PostgreSQL, OpenAI, Gemini, Claude* — Austin, TX

- Developed a full-stack AI feedback platform for programming courses, integrating code editing, file uploads, submissions, and AI chat feedback for debugging, algorithm complexity, implementation ideas, and test-case reasoning.
- Designed assignment-level AI controls so instructors can configure prompt templates, permitted context sources, model providers, request limits, and cooldown policies for course-specific feedback.
- Implemented context-filtering pipelines that selectively pass assignment descriptions, code, uploaded files, tests, execution results, and output to OpenAI, Gemini, Claude, or local models.
- Analyzed Docker-based autograder bottlenecks and scoped async grading with job status tracking, timing metrics, timeout handling, and isolated submission/result paths.

### Echo – Interactive AI Voice Simulation System — Jun. 2026 – Present

*University of Texas at Austin | Mentor: Prof. Bo Xie | Next.js, TypeScript, OpenAI API* — Austin, TX

- Built an event-driven voice simulation system that generates context-aware responses from user pauses, errors, skipped fields, and task switching using session state, trigger rules, browser TTS, and OpenAI APIs.
- Designed cooldown, interval, and event-priority logic to prevent repetitive playback and keep generated responses tied to task state and interaction timing.
- Modularized prompt templates and event-trigger logic so new voice events and task types can be added without changing the core session controller.

### Enterprise Risk Analyst Intern — May 2025 – Dec. 2025

*Toyota Industries Commercial Finance | Python, SQL, Azure, Power BI* — Dallas, TX

- Integrated PayNet inquiries and internal loan servicing data through SQL Server ETL, then engineered borrower, contract, and asset features including LTV, tenure, inquiry patterns, vehicle segments, and servicing attributes.
- Trained Logistic Regression and Random Forest credit-risk models, validating ROC/AUC, confusion matrices, and feature importance to achieve **AUC > 0.82** and improve high-risk case identification by **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard refinements.

### FinChat – AI-Powered Financial Analysis System — Jan. 2023 – May 2024

*Southern Methodist University | Python, FinGPT, LangChain, CrewAI* — Dallas, TX (Remote Hybrid)

- Built LLM workflows for financial report analysis, extracting and processing 100+ SEC filings from EDGAR into structured data for metric extraction, trend detection, and company-level comparison.
- Integrated FinGPT, LangChain, and CrewAI components for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time.
- Connected filing preprocessing, retrieval, metric extraction, and financial-question workflows into a reusable prototype for comparing companies across reporting periods.

### Data Scientist Assistant Intern – Image Restoration — Jul. 2023 – Sep. 2023

*Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN* — Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, split 3.4k images into train/validation/test sets, and achieved **80% accuracy** across 5 image categories.
- Integrated GFPGAN APIs for automated face restoration and color correction, supporting model-comparison reporting across restoration methods.

## Research Projects and Publications

### Researcher & TA – Biological Data Analytics — Jan. 2024 – Dec. 2024

*University of Texas at Austin | R, Linux, AWS, Docker* — Austin, TX

- Processed **250GB** of RNA-seq data with Salmon, RSEM, Trinity, and samtools, achieving a **98.7% mapping rate** across reproducible R/Linux pipelines.
- Performed Voom/Limma differential-expression analysis and identified 48 key genes with permutation-based FDR across time-series biological samples.
- Automated normalization, PCA, and volcano-plot workflows in R/ggplot2, improving interpretation efficiency for reproducible research outputs.

### Neuroengineering Lab – Machine Learning Research Assistant — Jan. 2023 – May 2024

*Python, MATLAB, MNE, RF, SVM, KNN, XGBoost* — Remote

- Processed **35GB** of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, statistical testing, and emotion-recognition model evaluation with RF, SVM, KNN, and XGBoost.
- Applied statistical tests and feature-extraction methods to evaluate cognitive impairment patterns and emotion-recognition signals across experimental conditions.
- Evaluated model behavior across experimental settings, contributing quantitative analysis used in peer-reviewed neuroscience and EEG/fNIRS publications.